

**1. Draw and Explain the NIST Cloud Reference Model (13 Marks)**

**Introduction**

The **NIST (National Institute of Standards and Technology) Cloud Reference Model** provides a common framework to understand cloud computing architecture, stakeholders, and their interactions. It helps in standardizing cloud services and ensuring interoperability, security, and portability.

**Main Actors in NIST Cloud Reference Model**

1. **Cloud Consumer**

- * End user or organization that uses cloud services
- * Requests services through web interfaces or APIs
- * Responsible for service usage and payment

2. **Cloud Provider**

- * Entity that offers cloud services
- * Manages infrastructure, platforms, and applications
- * Ensures availability, scalability, and security

3. **Cloud Broker**

- * Acts as an intermediary between provider and consumer
- * Performs service aggregation, integration, and cost optimization
- * Reduces complexity for consumers

4. **Cloud Auditor**

- * Independent entity
- * Evaluates security, performance, and compliance
- * Ensures SLA adherence and regulatory compliance

5. **Cloud Carrier**

- * Provides network connectivity
- * Transfers cloud services from provider to consumer
- * Ensures reliable communication

Explanation

The cloud provider delivers services such as **IaaS, PaaS, and SaaS**. The consumer accesses these services via the carrier. Brokers optimize services, while auditors ensure transparency and security.

(Draw diagram showing all five actors with arrows)

2. Compare Cloud Deployment Model and Cloud Service Model (13 Marks)

Introduction

Cloud computing models are classified into **deployment models** and **service models**, each serving a distinct purpose in cloud architecture.

**Cloud Deployment Models

Deployment models define **how and where** the cloud infrastructure is deployed.

- * **Public Cloud** – Shared infrastructure, cost-effective
- * **Private Cloud** – Dedicated infrastructure, high security
- * **Hybrid Cloud** – Combination of public and private
- * **Community Cloud** – Shared by similar organizations

**Cloud Service Models

Service models define **what type of service** is delivered to the user.

- * **IaaS** – Provides virtualized hardware resources
- * **PaaS** – Provides development platforms
- * **SaaS** – Provides ready-to-use applications

**Comparison Table

Deployment Model	Service Model
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Focuses on infrastructure ownership	Focuses on service abstraction
Defines access control	Defines user responsibility
Determines data location	Determines control level
Concerned with security boundaries	Concerned with service management

**3. Mention the Levels of Virtualization and Explain Its Implementation Details (13 Marks)**

Virtualization enables efficient utilization of computing resources by abstracting physical hardware into virtual components.

**Levels of Virtualization**

**1. Instruction Set Architecture (ISA) Level**

- * Translates source instruction set to target hardware
- * Used in cross-platform emulation

**2. Hardware Level Virtualization**

- * Hypervisor manages physical resources
- * Allows multiple operating systems
- * Common in cloud data centers

**3. Operating System Level Virtualization**

- * Uses a single OS kernel
- * Creates isolated containers
- * Lightweight and fast

**4. Library Level Virtualization**

- * System libraries are virtualized
- * Improves application portability

**5. Application Level Virtualization**

- * Applications run in isolated virtual environments
- * Example: Java Virtual Machine

**Implementation Details**

Each level differs in abstraction, performance, and isolation, allowing flexible deployment based on requirements.

**4. What is Hypervisor? Draw and Explain the Working of Type I Hypervisor (13 Marks)**

**Introduction**

A **hypervisor** is a core component of virtualization that enables multiple operating systems to run on a single physical machine.

**Definition of Hypervisor**

A hypervisor is a software or firmware layer that manages virtual machines by allocating hardware resources such as CPU, memory, and storage.

**Type I Hypervisor (Bare-Metal Hypervisor)**

**Architecture**

- * Installed directly on physical hardware
- * No host OS required
- * Sits between hardware and virtual machines

**Working**

1. Directly accesses hardware resources
2. Creates and manages virtual machines
3. Allocates CPU time and memory dynamically
4. Ensures isolation between VMs
5. Handles VM scheduling and execution

**Advantages**

- * High performance
- * Strong security
- * High scalability

**Examples**

- * VMware ESXi
- * Microsoft Hyper-V
- * Xen

Type-2 Hypervisor

Architecture

- Physical Hardware
- Host Operating System (Windows / Linux / macOS)
- Multiple Virtual Machines (Guest OS + Applications)

(In exam: draw a layered diagram showing Host OS between hardware and hypervisor)

Working of Type II Hypervisor

- The host operating system is first installed on the physical hardware.
- The Type II hypervisor is installed as an application on the host OS.
- When a virtual machine is created, the hypervisor requests resources (CPU, RAM, storage) from the host OS.
- The host OS communicates with the physical hardware using device drivers.
- The hypervisor allocates virtual hardware to each guest OS.
- Multiple guest operating systems run simultaneously but depend on the host OS for execution.
- The hypervisor ensures isolation between virtual machines.

Advantages

Easy to install and configure

Suitable for desktop and personal use

No need for dedicated hardware

Good for testing and development

Limitations

Lower performance compared to Type I

Less secure due to dependency on host OS

Not suitable for large-scale cloud environments

Examples

VMware Workstation

Oracle VirtualBox

Parallels Desktop